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ORIGINAL PAPER

Crude oil price fluctuation forecasting incorporating news sentiment based on improved sentiment lexicon

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Abstract
To effectively and accurately capture future fluctuations in crude oil prices from massive news data, this study proposes a forecasting method based on an improved sentiment lexicon and integrating news sentiment. Firstly, a rule-based approach was proposed to obtain a news collection containing price fluctuation information. Secondly, an automatic algorithm for constructing an improved sentiment lexicon for oil price forecasting was proposed. The algorithm first updates the sentiment polarity of some words in the Loughran&McDonald (LM) financial lexicon, combined with the LM oil price lexicon, to establish an expanded seed lexicon (LM-S), and then extends LM-S to form the final domain specific sentiment lexicon. Thirdly, a news classification model was constructed, dividing news into eight categories and extracting emotional features comprehensively through four types of emotional indicators. Finally, by integrating WTI crude oil prices, sentiment indicators, and financial indicators, and using feature screening and lag calculation, LSTM, BP, and GRU models are used to predict future price fluctuations of crude oil. In the experiment, directional accuracy, MAPE, and the newly proposed directional absolute error rate were used as evaluation indicators. Shapley values were used to explain the contribution of each feature to the forecasting results.

Keywords Crude oil price fluctuation forecasting · Oil price forecasting sentiment lexicon · Sentiment analysis · Text classification · Online news

1 Introduction

Accurately predicting crude oil price fluctuations that is, the direction changes of crude oil prices is crucial because they have a significant impact on countries and companies. As noted by Zhang et al. (2015) crude oil price fluctuations affect national economic policies and economic trends, as well as the investment decisions of individual companies and determine their profits and losses. While conventional factors such as crude oil supply and demand (Hagen 1994; Stevens 1995) have historically influenced crude oil prices, unconventional factors such as wars and political events (Tang et al. 2015) have also played a role. Predicting fluctuations in crude oil prices is a significant challenge due to the impact of uncertainties.

Many scholars have used financial factors as influencing factors to predict crude oil prices. However, short-term changes in oil prices are highly related to major events (Zhang et al. 2009). Financial factors alone may not reflect the influence of these unconventional factors in a timely and accurate manner. The news published on the internet can reflect the dynamics of events related to the oil price market in a timely and comprehensive manner. Therefore, capturing oil price market changes from the news has become a possibility (Sadik et al. 2019). Scholars have incorporated news sentiment analysis methods into oil price forecasting. They have mined and analyzed the sentiment information of oil prices contained in the news and quantified them (Li et al. 2018, 2021; Bai et al. 2022). However, most of these studies use a generic sentiment lexicon that is not fully applicable to predicting oil price fluctuations (Zhao et al. 2023). In addition, some studies have utilized news categories obtained

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by clustering news articles with the use of clustering methods. However, since clustering is an unsupervised learning method, although there are methods such as Silhouette to evaluate the clustering results, clustering faces several problems. Firstly, if the number of clusters is not chosen properly, some categories may be ignored or subdivided, whereas the categories of supervised learning methods are sure; and secondly, clustering methods are more sensitive to noise and isolated points. Especially datasets from the Internet have a higher percentage of noise points compared to datasets that exist. Noisy points interfere with points that are close or similar to them, and if there are more noise points, they may also be divided into separate clusters, which makes the clustering effect worse.

To effectively forecast the direction changes and values of crude oil future prices, this paper proposes a novel algorithm to forecast crude oil price fluctuations incorporating news sentiment based on an improved sentiment lexicon. Firstly, a collection of news containing price fluctuations information is obtained using the developed rule-based approach. Afterward, a new improved oil price forecasting sentiment lexicon is developed by establishing the seed sentiment lexicon Loughran&McDonald seed lexicon (LM-S) based on Loughran&McDonald (LM) financial lexicon (Loughran and McDonald 2011) and the LM oil price lexicon (Loughran et al. 2019), and further expanding the LM-S lexicon for accurate news sentiment calculation. Then, the news is classified into eight categories, and four types of sentiment indicators are used to extract the news sentiment features for oil price forecasting. Finally, a forecasting algorithm based on multiple deep learning models is constructed to forecast crude oil future price fluctuations using WTI crude oil prices, as well as the selected news sentiment features and financial features, and SHAP values are used to interpret the forecasting results. To evaluate the forecasting performance more accurately, new indicators that include directional and absolute error metrics are further developed.

The main innovations and contributions can be summarized as follows:

(1) This study proposes a novel algorithm to forecast crude oil price fluctuation incorporating news sentiment based on the developed improved oil price forecasting sentiment lexicon. It can effectively forecast the direction changes and values of crude oil future prices and provide explanatory information for the results.
(2) A novel automatic approach to build an improved oil price forecasting sentiment lexicon is proposed for news sentiment calculation. It alleviates the burden of manual sentiment labeling and makes the established oil price forecasting sentiment lexicon more comprehensive and accurate. The approach can also be flexibly applied to other domain sentiment lexicons.
(3) A new type of sentiment indicator is designed to measure the news sentiment comprehensively. It helps to extract the critical features of news sentiment for more accurate oil price fluctuation forecasting.
(4) The experimental results along with comparative analysis demonstrate the superior performance of the proposed method and verify the effectiveness of the established oil sentiment lexicon and the sentiment indicators.

The following sections of the paper are structured as follows: Section 2 provides a brief literature review of oil price forecasting techniques and sentiment analysis theory. Section 3 introduces the proposed forecasting method. Section 4 presents the experimental results and analysis. Finally, the conclusion and outlook of this study are presented in Section 5.

2 Literature review

Currently, most studies on oil price forecasting use traditional statistical methods, including the autoregressive integrated moving average model (ARIMA) (Mohammadi and Su 2010; Xiang and Zhuang 2013) and the generalized autoregressive conditional heteroskedasticity (GARCH) (Hou and Suardi 2012), which rely heavily on historical oil prices for forecasting. However, the predicted results of these studies often depend too much on the price fluctuations in the past. Some scholars not only consider the oil price itself but also financial factors and use various machine learning methods to predict oil prices, such as the decision tree (Hacer et al. 2015; Gumus and Kiran 2017), the neural network (Ma 2010; Movagharnejad et al. 2011), and other methods, achieving better forecasting results. However, the existing research has not typically considered the influence of unconventional factors on oil price forecasting (Dong et al. 2022b, a).

With the advancement of big data and text mining technology (Li et al. 2024), utilizing potential information from text data provides new insights to strengthen oil price forecasting (Jawadi et al. 2024; Jiang et al. 2025). The study by Wex et al., (2013) indicates that online news messages have a powerful predictive ability for oil price forecasting. Yu et al., (2005) proposed a knowledge-based method RSTM for predicting crude oil price trends. Wang et al., (2004) developed a hybrid artificial intelligence forecasting framework based on web text mining techniques to predict crude oil prices. Additionally, sentiment analysis techniques are widely used in news text mining, and many scholars construct news sentiment features by analyzing news sentiment to predict market fluctuations. Li et al., (2018) proposed to extract news features by deep learning techniques and predict crude oil prices with financial features. Li et al., (2021) built a crude oil prices forecasting model VMD-BILSTM considering investor sentiment indicators. (Wu et al. 2021a, b) used word embedding pre-training models and deep learning methods for automatic feature extraction from news texts, and Bai et al., (2022) proposed new textual metrics for characterization of short and sparse textual news headlines for oil price forecasting. However, in the text analysis usages of oil price forecasting, most scholars use the same research method as other text analysis tasks (Dong et al. 2025), the current research for the calculation of news sentiment value relies on existing sentiment analysis tools, or based on non-specialized fields of data, general text analysis or sentiment analysis methods and models, the analysis process lacks the feature words of the petroleum field, so its sentiment analysis of the news is not fully applicable in the field of oil price forecasting.

In addition to sentiment analysis, researchers have also recognized that news topics and their discussion degree can reflect the fluctuation of oil prices. They have used topic models to analyze the topics that influence market prices. For example, Nguyen et al., (2015) proposed to select both topics and market sentiments using a joint model. Li et al., (2018) used the Latent Dirichlet Allocation (LDA) method to differentiate various online topics. Bai et al., (2022) analyzed news themes by the SeaNMF method to predict oil prices. However, in these studies, news categories were obtained using clustering methods, making it difficult to evaluate the accuracy of their classification results and to set the number of clusters correctly.

In the field of crude oil price forecasting, investors are more interested in predicting oil price fluctuations rather than numerical price values because they want to use models to predict future fluctuations to guide further decisions. Investors are not as sensitive to the absolute value of oil prices as they are to the fluctuations. An increase in the price of oil from \$66 to \$70 a barrel is the same as an increase in the absolute value of the price of oil from \$16 to \$20 a barrel, but the latter is more fluctuating and more profitable for investors, and thus the fluctuations affect investors' decisions more than the absolute value of the price of oil. Therefore, some scholars have adopted heteroscedasticity adjusted mean squared error (HMSE) and heteroscedasticity adjusted mean squared error (HMAE) as evaluation criteria for mean squared error (MSE) and mean absolute error (MAE) forecasting fluctuations (Dong et al. 2021; Li et al. 2021; Niu et al. 2021). However, these criteria only measure the trend of fluctuations between the predicted and true values in the current period and cannot reflect the effect of oil price fluctuation on the previous period. Jiang et al., (2022) improved upon this by calculating the directional accuracy, which only considers the directional accuracy but not the numerical accuracy.

3 Proposed method

The overall framework of the proposed crude oil price fluctuation forecasting method is depicted in Fig. 1.

Firstly, construct the novel improved oil price forecasting sentiment lexicon. A collection of news containing price fluctuations information is obtained using a rule-based approach. The sentiment polarity of a subset of words in the Loughran&McDonald (LM) financial lexicon(Loughran and McDonald 2011) is calculated and combined with the LM oil price lexicon to create an expanded seed lexicon, Loughran&McDonald seed lexicon (LM-S). To further enrich the oil price forecasting sentiment lexicon, the sentiment polarity of other words in the LM financial lexicon is calculated using the method proposed by Jiang et al., (2019), which includes PMI and Word2Vec.

Secondly, extract news sentiment features for oil price forecasting. The news is classified into eight categories, and four types of sentiment indicators are designed. The values of each sentiment indicator under each news category are calculated using the improved oil price forecasting sentiment lexicon.

Finally, develop the forecasting model for crude oil price and evaluate the performance. The feature selection and lag period calculation are performed on WTI crude oil prices, financial features and news sentiment features. New forecasting evaluation criteria are designed based on the forecasting values and direction. On this basis, crude oil prices are predicted using long short-term memory (LSTM) (Song et al. 2020), backpropagation (BP) (Li et al. 2012), and gate recurrent unit (GRU) methods (Salem 2021), and the forecasting results were interpreted using Shapley (SHAP) (Kim and Kim 2022) values.

3.1 Improved oil price forecasting sentiment lexicon construction

The LM financial lexicon (Loughran and McDonald 2011) contains a large number of words with sentiment semantics. The LM oil price lexicon (Loughran et al. 2019) contains a limited number of sentiment words and cannot fully capture the factors influencing oil prices. However, the polarity of the words in the LM financial lexicon may not be directly applicable to oil price forecasting. Therefore, in expanding the lexicon, the polarity of the words in the LM financial lexicon needs to be updated and combined with the LM oil price lexicon to create a comprehensive sentiment lexicon.

In this section, we first obtain news sets with price fluctuation labels based on proposed rules. Next, we update

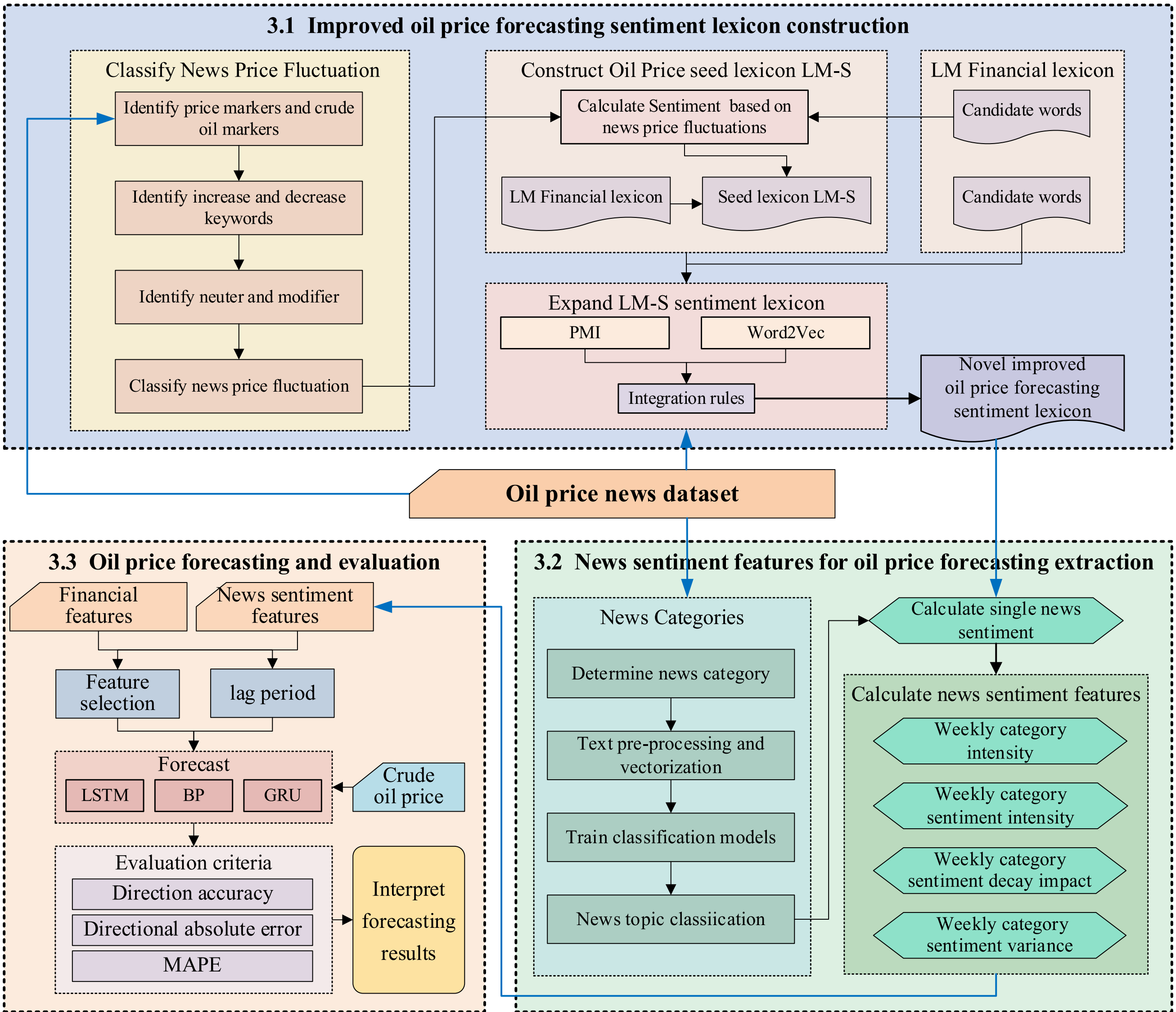

Fig. 1 Overall framework

the polarity of a subset of words in the LM financial lexicon and combine it with the LM oil price lexicon to obtain the expanded seed lexicon LM-S. Then, we update the polarity of other words in the LM financial lexicon using LM-S and the news through the inter-word-based co-occurrence probability method and the semantic similarity measure method. This results in a more comprehensive sentiment lexicon for oil price forecasting. The framework for enriching the oil price forecasting sentiment lexicon is illustrated in Fig. 2.

3.1.1 Rule-based determination of oil price label of news

News headlines reflect the key content of news and can serve as the primary basis for classifying news price fluctuations. Therefore, we match keywords to news headlines to determine news price fluctuations. Initially, we match news headlines with price identifiers to filter news related to prices. Next, we match the news with crude oil identifiers to filter the news that includes crude oil price information. Subsequently, we match the news headlines with keywords related to oil price increases or decreases. If such keywords are present, the news headlines are classified into price fluctuations. If there are no such keywords, the news headlines are classified by identifying modifiers and neutral words. Finally, the result is used as a news fluctuation category (Dong et al. 2023).

The steps of the proposed rule-based news price label determination method are shown in Table 1. The news headlines are classified by the following steps:

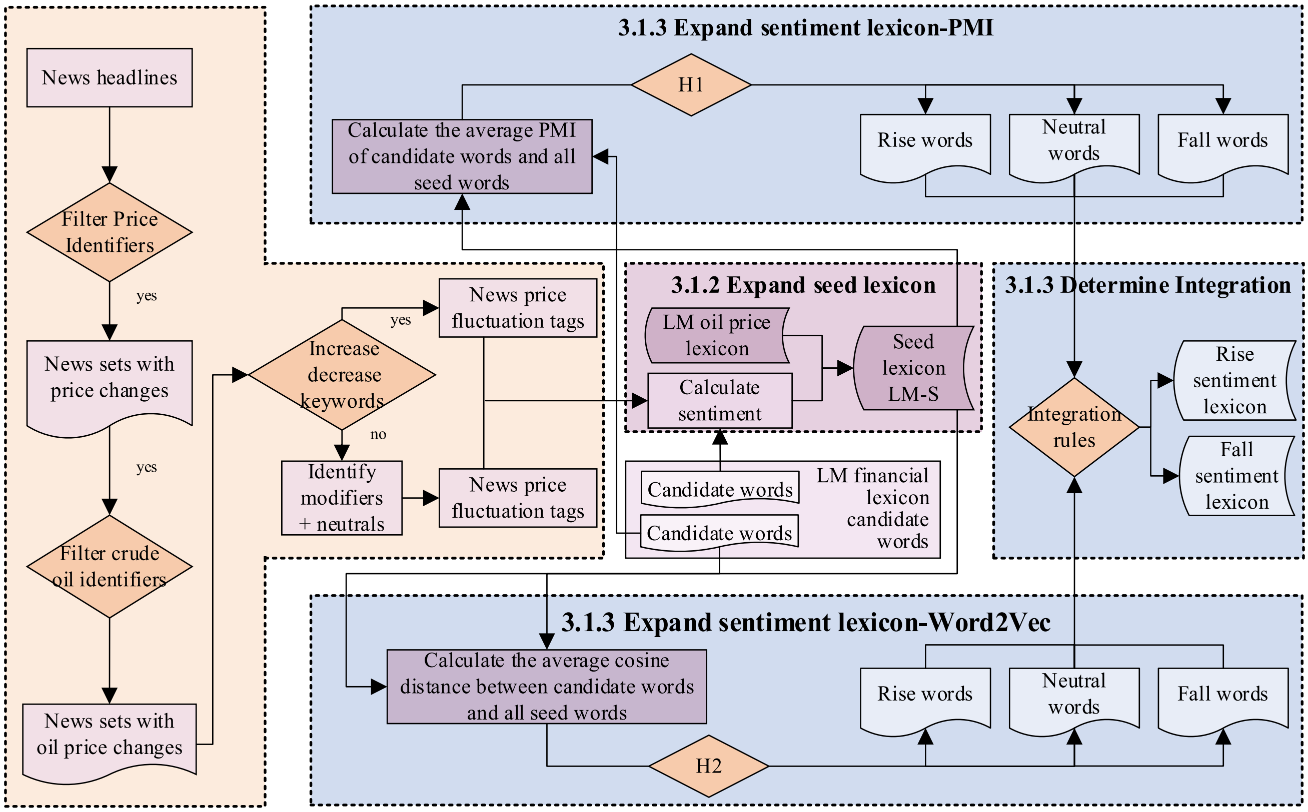

Fig. 2 Enriching the oil price forecasting sentiment lexicon

Firstly, match each word in each news headline with each word in *Price_list*, which includes price identifiers; if at least one word in a news headline can match with price identifiers, this news headline is related to price rise and fall; if there is no word that can be matched, it means this news headline is not related to price rise and fall, and the sample will be excluded. Which "*Price_list*" is a collection of core terms used to identify content directly related to crude oil prices in news, including terms such as "oil prices" and "price change".

Secondly, each word in *Price_news* is matched with each word in *Oil_list*, which includes oil identifiers, if at least one word in a news headline can match with oil identifiers, it means that this news headline is related to crude oil, and the sample will be added to the candidate news list *Oil_news*; if there is no word that can be matched, it means that this news headline is not related to crude oil, and the sample will be excluded, so that we can get the news collection *Oil_news* related to the rise and fall of international crude oil price.

Then, the words in each news headline in the set are compared with the keywords in *Rise_lis* t and *Fall_list*. "*Rise_list*" is a collection of words describing the upward trend of prices, including words such as "rise" and "increase", while "*Fall_list*" is a collection of words describing the downward trend of prices, including words such as "drop" and "fall". If at least one word in a news headline can only match the keywords in the *Rise_list*, then this news is related to rising oil prices, and this news will be classified as *Rising_label*. Similarly, if at least one word in the headline of a news can and can only match the keywords in the *Fall_list*, then this news headline is related to falling oil prices, and this news will be classified as a *Drop_label*. If a piece of news does not match the flag word in both keyword lists, or if it matches the flag word in both keyword lists, then this news needs to be further categorized and added to *Mid_news*.

For the *Mid_news* collection, the words in each news headline are first matched with the neutral word in *Mid_list*, if no neutral word is matched, the sample will be excluded. If the neutral word is matched, match the modifier of the neutral word with the positive modifier in the *Pos_list* and the negative modifier in the *Neg_list*. If the modifier can and can only be matched with the positive modifier, then this news is related to the rising oil price, and this news will be categorized as *Rising_label*. If the modifier can and can only match the negative modifier, then this news is related to the fall of oil price and this news will be categorized as *Drop_label*, while other pieces of news are considered to contain no price fluctuation information.

Table 1 The rule-based news price label determination method

Algorithm 1 Rule - based news price label determination

Input: $Headlines = \{h_1, h_2 \ldots h_n\}$, the set of news headlines, $h_i = \{w_{i1}, w_{i2} \ldots w_{im}\}$, the set of words in the headline; $Price\_list = \{w\_P_1, w\_P_2 \ldots w\_P_a\}$, the set of words denoting price; $Oil\_list = \{w\_O_1, w\_O_2 \ldots w\_O_B\}$, the set of words denoting oil; $Rise\_list = \{w\_R_1, w\_R_2 \ldots w\_R_C\}$, the set of words denoting a rise in price; $Fall\_list = \{w\_F_1, w\_F_2 \ldots w\_F_D\}$, the set of words denoting a fall in price; $Mid\_list = \{w\_M_1, w\_M_2 \ldots w\_M_E\}$, the set of neuter words needs to be determined; $Pos\_list = \{w\_P_1, w\_P_2 \ldots w\_P_F\}$, the set of positive modifiers; $Neg\_list = \{w\_N_1, w\_N_2 \ldots w\_N_G\}$, the set of negative modifiers.

Output: $Rise\_news$, the set of news that indicate oil prices will rise; $Fall\_news$, the set of news that indicate oil prices will fall.

for each h_i in $Headlines$ **do**
  if $h_i \cap Price\_List$ is not null **then**
    Add h_i to $Price\_news$(news about price)
  else
    Remove h_i from $Headlines$
  end if
end for
for each h_i in Price_news do
  if $h_i \cap Oil\_List$ is not null **then**
    Add h_i to Oil_news(news about oil)
  else
    Remove h_i from $Price\_news$
  end if
end for
for each h_i in Oil_news **do**
  if $\{h_i \cap Rise\_List\}$ is not null and $\{(h_i \cap Rise\_List)(h_i \cap FallList)\}$ is null **then**
    Add h_i to $Rise\_news$
  else if $\{h_i \cap Fall\_List\}$ is not null and $\{(h_i \cap Rise\_List)(h_i \cap Fall\_List)\}$ is null **then**
    Add h_i to $Fall\_news$
  else
    Add h_i to Mid_news(the news needs to be identified through word matching)
  end if
end for
for each h_i in Mid_news **do**
  if $\{h_i \cap Mid\_List\}$ is not null **then**
    if $\{h_i \cap Pos\_List\}$ is not null and $\{(h_i \cap Pos\_List) \cap (h_i \cap Neg\_List)\}$ is null **then**
      Add h_i to $Rising\_label$
    else if $\{h_i \cap Neg\_List\}$ is not null and $\{(h_i \cap Pos\_List) \cap (h_i \cap Neg\_List)\}$ is null **then**
      Add h_i to $Drop\_label$
    else
      Remove h_i from Mid_news
    end if
  else
    Remove h_i from Mid_news
  end if
end for
Matching news to headlines
return $Rise\_news$, $Fall\_news$

The classification results in two collections of news containing information about price fluctuations.

3.1.2 Expanded seed lexicon LM-S construction

After obtaining news sets containing price fluctuation information, this section updates the sentiment polarity of a subset of words in the LM financial lexicon and combines it with the LM oil price lexicon to construct the expanded seed lexicon LM-S.

Initially, we calculate the frequencies of the words in the LM financial lexicon that appear in news with price fluctuations. A word is added to the rising seed lexicon if the difference between the frequency of the word appearing in the news with the price tag "rise" and "fall" is greater than the threshold H. If the difference between the frequency of the word appearing in the news with the price tag "decreasing" and "increasing" is higher than the threshold H, then the word is added to the falling seed lexicon. Regarding the threshold setting, if H were set too high, the lexicon would be too small, risking the omission of important sentiment information; conversely, if H were too low, the number of selected sentiment words would be excessively large, introducing terms that are weakly related to crude oil price fluctuations. In practical applications, it can be adjusted according to specific research scenarios and data characteristics to adapt to different requirements for lexicon scale and precision. If the difference in occurrence frequency is insignificant, the word is retained for polarity determination in the next sentiment lexicon expansion step. The obtained rising words (e.g., supportive) and falling words (e.g., unprofitable) are then combined with the LM oil price lexicon to create the expanded seed lexicon LM-S. The specific algorithm is presented in Table 2.

Table 2 Expand seed lexicon

Algorithm 2 Expand seed lexicon

Input: $Rise\_news = \{n\_R_1, n\_R_2 \ldots n\_R_n\}$, the set of news that indicate oil prices will rise; $Fall\_news = \{n\_F_1, n\_F_2 \ldots n\_F_m\}$, the set of news that indicate oil prices will fall; $Candidate\_words = \{w\_C_1, w\_C_2 \ldots w\_C_A\}$, the set of candidate words in the LM Financial Lexicon; $Rise\_words = \{w\_R_1, w\_R_2 \ldots w\_R_B\}$, the set of words indicating rising oil prices from the LM Oil Price Lexicon; $Fall\_words = \{w\_F_1, w\_F_2 \ldots w\_F_C\}$, the set of words indicating falling oil prices from the LM Oil Price Lexicon

Output: LM_S, the set of new seed word; C_W, the set of candidate words in the LM Financial Lexicon for further determination

Let $Fre\_Rise(w\_C_i, Rise\_news)$ denote the frequency of the word w_C_i in the news that indicate oil prices will rise
Let $Fre\_Fall(w\_C_i, Fall\_news)$ denote the frequency of the word w_C_i in the news that indicate oil prices will fall
Let H denote the threshold to the frequency gap
for each w_C_i in $Candidate\_words$ **do**
  if $Fre\_Rise(w\_C_i, Rise\_news) > Fre\_Fall(w\_C_i, Fall\_news)$ and $Fre\_Rise(w\_C_i, Rise\_news) - Fre\_Fall(w\_C_i, Fall\_news) > H$ **then**
    move w_i from $Candidate\_words$ to $Rise\_words$
  else if $Fre\_Fall(w\_C_i, Fall\_news) > Fre\_Rise(w\_C_i, Rise\_news)$ and $Fre\_Fall(w\_C_i, Fall\_news) - Fre\_Rise(w\_C_i, Rise\_news) > H$ then
    move w_i from $Candidate\_words$ to $Fall\_words$
  else
    move w_i from $Candidate\_words$ to C_W
  end if
end for
$Rise\_words$ and $Fall\_words$ combined as LM_S
return LM_S, C_W

3.1.3 Expansion of the sentiment lexicon

The LM-S seed lexicon serves as the core foundation for sentiment lexicon expansion. Specifically, LM-S filters out core positive and negative seed words through frequency differences, providing initial polarity annotations for the semantic expansion tasks based on PMI and Word2Vec in this section (Jiang et al. 2019). This forms a complete process of "first obtaining the expanded seed lexicon, then constructing a domain-specific lexicon based on this seed lexicon."

After obtaining the expanded seed lexicon LM-S, we calculate the polarity of the remaining words in the LM financial lexicon. We use PMI in the inter-word co-occurrence probability method and Word2Vec in the semantic similarity method to obtain the polarity, and then merge the two polarities using integration rules to derive the final polarity. Additionally, the candidate words in the LM lexicon have already been preprocessed, so it's unnecessary to preprocess these words.

The PMI-based method for determining candidate word polarity involves calculating the textual co-occurrence between two words in a corpus of news about oil prices. This measures the proximity between a new word with an unknown oil price forecasting sentiment polarity and a seed word with a known oil price forecasting sentiment polarity, thus determining the oil price forecasting sentiment category to which the new word belongs (Church and Hanks 1989).

$$PMI\left(w_1, w_2\right) = log_2 \frac{p(w_1, w_2)}{p\left(w_1\right)p\left(w_2\right)} \tag{1}$$

$$sentiScore(word) = \frac{1}{N_{rise}} \sum_{i=1}^{N_{rise}} PMI\left(word, riseSeed_i\right) - \frac{1}{N_{fall}} \sum_{j=1}^{N_{fall}} PMI(word, fallSeed_j) \tag{2}$$

where w_1 and w_2 represent two words, $p(w_1)$ denotes the probability of w_1 occurring in the corpus of news about oil prices, $p(w_2)$ denotes the probability of w_2 occurring in the corpus of news about oil prices, $p(w_1, w_2)$ denotes the probability of w_1 and w_2 occurring in the corpus of news about oil prices simultaneously, and $PMI\left(w_1, w_2\right)$ denotes the association degree of w_1 and w_2. *riseSeed* and *fallSeed* denote the set of seed words representing a rise in oil prices and the set of seed words representing a fall in oil prices, respectively. A larger value of $sentiScore(word)$ indicates a stronger polarity of the candidate word representing a rise in oil prices, while a smaller value indicates a stronger polarity of the candidate word representing a fall in oil prices.

The Word2Vec-based sentiment lexicon expansion method evaluates the degree of semantic relationship between words by calculating the cosine similarity from the mapped vectors. Initially, Word2Vec (Ma and Zhang 2015) maps two words or phrases, w_1 and w_2 into n-dimensional vectors, $w_1 = x_1, x_2, \cdots, x_n$, and $w_2 = y_1, y_2, \cdots, y_n$. Next, using the cosine formula, we calculate the average semantic similarity between the candidate words and all the oil price increase and decrease seed words in the seed lexicon. Subsequently, we derive the sentiment scores of the candidate words for oil price forecasting (Lustosa Filho et al. 2018).

$$cos(w_1, w_2) = \frac{\sum_{i=1}^{n} x_i y_i}{\sqrt{\sum_{i=1}^{n} x_i^2}\sqrt{\sum_{i=1}^{n} y_i^2}} \tag{3}$$

$$sentiScore(word) = \frac{1}{N_{rise}} \sum_{i=1}^{N_{rise}} cos\left(word, riseSeed_i\right) - \frac{1}{N_{fall}} \sum_{j=1}^{N_{fall}} cos(word, fallSeed_j) \tag{4}$$

where $sentiScore(word)$ is the similarity between *word* and words from $riseSeed_i$ and $fallSeed_j$, and a larger value of $sentiScore(word)$ indicates the stronger polarity of the candidate word representing an increase in oil prices, while a smaller value indicates a stronger polarity of the candidate word representing a decrease in oil prices.

In this study, we use the PMI-based sentiment polarity discrimination model and the Word2Vec-based sentiment polarity discrimination model as two base classifiers, C1 and C2. We obtain two sets of classification results of candidate words from the corresponding models: words representing a falling polarity of oil prices, neutral words, and words representing a rising polarity of oil prices, denoted as $y = y_{-1}, y_0, y_1$. Assuming the probability of the three types of words occurring is $P(y) = P\left(y_{-1}\right), P\left(y_0\right), P(y_1)$, we fuse the results of the two base classifiers using the improved voting rule, and adjusted majority vote rule (AMVR). The fusing function is presented in Eq. (5) (Jiang et al. 2019).

$$y = max\left\{ \sum_{C_1,C_2} \frac{y_{-1}}{P(y_{-1})}, \sum_{C_1,C_2} \frac{y_0}{P(y_0)}, \sum_{C_1,C_2} \frac{y_1}{P(y_1)} \right\} \tag{5}$$

where $P\left(y_{-1}\right) < P\left(y_0\right), P\left(y_1\right) < P\left(y_0\right)$.

3.2 News sentiment features for oil price forecasting extraction

In this section, we first classify news into eight categories by analyzing the factors influencing oil price fluctuations and constructing a machine learning model to classify news into these categories. Next, we define four types of sentiment indicators. Finally, we calculate the values of the four sentiment indicators for each news category using the oil price forecasting sentiment lexicon. The process is presented in Fig. 3.

3.2.1 News category classification

The price of crude oil, an essential commodity, is determined by the forces of supply and demand. The demand for crude oil is relatively stable over a given period due to the relative stability of economic development (Miao et al. 2017). However, oil-producing countries can adjust their production through surplus capacity, making supply more susceptible to short-term price fluctuations. Notably, OPEC + and the United States have the strongest production flexibility (Su et al. 2020). In addition to supply and demand, research has shown a clear negative correlation between crude oil inventory and oil prices (Bu 2014). As a commodity priced in US dollars, the monetary policy of the US Federal Reserve significantly impacts crude oil prices (Zhao et al. 2022). Loose monetary policy by the Fed will drive nominal oil prices upward, and interference by managed funds' positions can also disrupt oil price movements. Geopolitically, sudden geopolitical conflicts can affect oil prices by altering supply and demand (Xu et al. 2021). Additionally, extreme weather events such as floods, heatwaves, and variations in weather patterns can disrupt the crude oil supply side. Simultaneously, global public health security events can influence crude oil demand. Therefore, we identify the news categories of OPEC + crude oil production, US crude oil production, crude oil inventory, crude oil demand, finance, extreme weather, pandemics, and geopolitical conflicts (Gkillas et al. 2022; Zhao 2022).

After determining the news categories, we construct a classification model to automatically classify the news. Since news articles are often long and contain multiple topics, we first divide them into paragraphs and manually classify each paragraph into the corresponding category. However, the number of news articles in different categories varies significantly, resulting in imbalanced data that may cause the classification model to be influenced by the majority of categories during the training process, thereby ignoring information from minority categories. To address this, we first check the balance of each

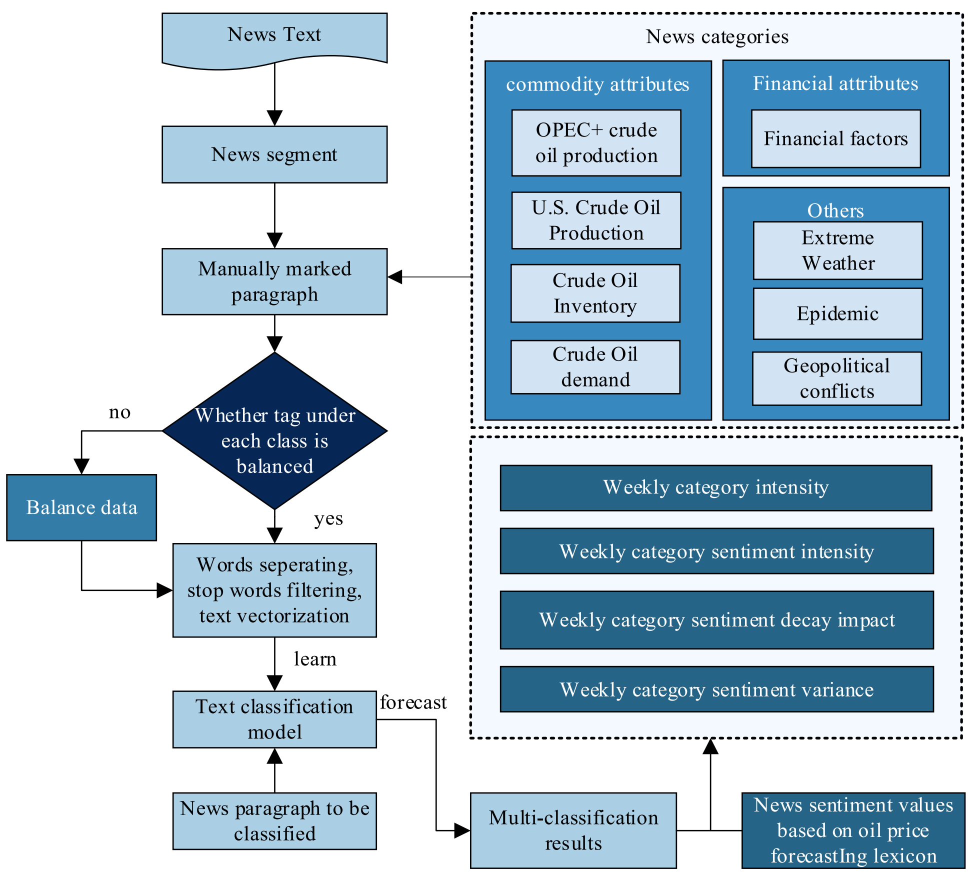

Fig. 3 Calculate the news sentiment feature

class and balance unbalanced ones (Li et al. 2021). Next, we preprocess the text of news paragraphs by separating words, filtering stop words, and vectorizing the text using word frequency-inverse document frequency (TF-IDF) (Luo et al. 2002). The classifiers are obtained by training models such as random forest (Breiman 2001), XGBoost (Chen and Guestrin 2016), and LightGBM (Ke et al. 2017). Finally, the classifier predicts the category to which the news paragraph belongs, and the categories of each news paragraph are combined to form the multi-categories of the news.

3.2.2 News sentiment features calculation

In this study, we calculate the sentiment value of news using semantic rules based on the oil price forecasting sentiment lexicon (Tianfang and Hongxun 2021) according to the following rules.

$$U_1 = \prod L_s \times sentiScore \tag{6}$$

$$U_2 = (\prod L_n \times (0.1 \times C_n - 1)^{C_n} + \prod L_s) \times sentiScore \tag{7}$$

$$News = \sum_j U_{1,j} + \sum_l U_{2,l} \tag{8}$$

In the equation, U_1 represents the sentiment value of phrases without negatives, U_2 represents the sentiment value of phrases that contain negatives, *sentiScore* represents the sentiment score of the sentiment word that signifies the change in oil price, L_s represents the degree value of degree adverbs modifying sentiment words, L_n represents the degree value of degree adverbs modifying negatives, C_n represents the number of negatives, and *News* represents the sentiment value of the oil price change for news.

Using the news categories and sentiment values of each news, we calculate the weekly category intensity, weekly category sentiment intensity, and weekly category sentiment decay impact (Bai et al. 2022). Additionally, we construct weekly category sentiment variance indicators to measure the degree of intra-weekly news variation.

(1) Weekly category intensity

A significant number of news items discussing oil prices are published weekly on internet platforms. As news items belong to different categories, the category intensity in different periods reflects the news activity level under that category in that period. The weekly category intensity is calculated using the following formula (Bai et al. 2022):

$$CW_{i,j} = \frac{N_{i,j}}{N_j} \tag{9}$$

$$CI_{i,t} = \frac{1}{N_t} \sum_{j=1}^{N_{i,t}} CW_{i,j} \tag{10}$$

where N_j is the number of categories of news j, $N_{i,j}$ is the tag of news j under category i. $N_{i,j}$ equals 1 if news j belongs to category i; otherwise, $N_{i,j}$ equals 0. CW_{ij} represents the weight of the j th news under the category i in week t. $N_{i,t}$ is the number of oil price news published under the i th category in week t, and $N_t i$ s the total number of news releases on oil prices during the week. $CI_{i,t}$ represents the category intensity of the i th category in week t.

(2) Weekly category sentiment intensity

The sentiment contained in the oil price news expecting oil prices to rise or fall will reflect the crowd's judgment of the market. The category sentiment intensity for the week is obtained by the following calculation (Bai et al. 2022).

$$CSI_{i,t} = \frac{1}{N_{i,t}} \sum_{k=1}^{N_{i,t}} News_{i,k,t} \tag{11}$$

where $News_{i,k,t}$ represents the sentiment value of the k th news item in category i in week t, $N_{i,t}$ represents the number of news items under category i in week t, and $CSI_{i,t}$ represents the sentiment intensity of category i in week t.

(3) Weekly category sentiment decay impact

There is a cumulative and decreasing effect of news on public sentiment, with current sentiment being influenced by previous news in addition to the current week's oil price news. It is therefore assumed that the effect of the news on public sentiment decreases exponentially over time. Based on this assumption, the weekly category sentiment intensity is constructed. The calculation formula is as follows (Bai et al. 2022):

$$CS\_DI_{i,t} = \sum_{l=1}^{t-1} e^{-\frac{t-l}{n}} CSI_{i,l} + CSI_{i,t} \tag{12}$$

where $CSI_{i,t}$ represents the average sentiment intensity under category i in week t, n represents the effect of news on prices in the next n weeks (i.e., news has an impact on prices in the next n weeks), $e^{-\frac{t-l}{n}} CSI_{i,l}$ represents the sentiment impact of the average sentiment intensity of category i in week l on week t, and $CS\_DI_{i,t}$ represents the sentiment impact of week t's category sentiment intensity for category i considering the effect of decay.

(4) Weekly category sentiment variance

The magnitude of the variance of the sentiment scores for news under each category over a week indicates how discrete the news is in terms of sentiment in that category.

The higher the sentiment variance value, the more discrete it is, indicating that news in that category is more divergent in sentiment towards the rise and fall of oil prices that week. Conversely, the lower the sentiment variance value, the less discrete it is, indicating that news in that category is more consistent in sentiment toward the rise and fall of oil prices that week. To calculate the sentiment variance for week t, we use the category sentiment intensity ($CI_{i,t}$) of each week's news release as a weighting factor, and define it as follows:

$$CSI\_V_{i,t} = CI_{i,t} \times \sum\nolimits_{j}^{n} \frac{\left(News_{i,k,t} - CSI_{i,t}\right)}{N_{i,t} - 1} \tag{13}$$

where $N_{i,t}$ denotes the number of news items published under category in week t. $News_{i,k,t}$ denotes the sentiment value of the j th news item under category i in week t. $CSI_{i,t}$ represents the mean sentiment intensity under category i in week t. $CI_{i,t}$ represents the mean sentiment intensity under category i in week t. $CSI\_V_{i,t}$ represents the sentiment variance of category i in week t.

3.3 Oil price forecasting and evaluation

In this section, the first step involves selecting features from all available financial and news sentiment features. Following this, the lags of the selected features are calculated using the VAR lag selection criterion. The next step involves constructing and training forecasting models using the selected features. Finally, the performance of the forecasting models is evaluated using relevant evaluation criteria.

3.3.1 Feature selection and lag period

The recursive feature elimination (RFE) method is a commonly used approach for feature selection (Li et al. 2018; Bai et al. 2022), which involves recursively reducing the size of the feature set to identify the optimal features. However, this method fails to account for the correlation between features during the recursive process, resulting in the inclusion of redundant features in subsequent calculations. To address this issue, we adopt a two-stage feature selection method based on feature ranking and approximate Markov gaps, known as the fast correlation-based filter solution-maximal information coefficient (FCBF-MIC) method (Sun et al. 2017; Pan et al. 2023; Feng et al. 2025). This method allows for the selection of features from both news sentiment feature indicators and financial indicators, considering both their relevance and redundancy. The framework of this method is shown in Fig. 4.

The first stage of our two-stage feature selection method involves correlation analysis, in which we use the symmetric uncertainty (SU) measure (Sosa-Cabrera et al. 2019) to assess the correlation between financial data, indicator data, and crude oil prices. We then use the feature ranking method to eliminate factors that are either irrelevant or weakly correlated, resulting in the identification of factor subsets. In the second stage, we perform redundancy analysis by calculating the maximum information coefficient (MIC) (Simon and Tibshirani 2014) between the subset of factors and applying the approximate Markov blanket method to eliminate redundant factors (Wang et al. 2004).

In this study, we utilized the vector autoregression (VAR) model (Ivanov and Kilian 2005) to calculate feature lag. This model represents each variable as a linear combination of its past values and those of other variables in the system, capturing the interdependencies among multiple time series. As each influencing factor series may affect the crude oil price series, we modeled the VAR model as a system of equations to analyze the relationships between all influencing factor series and the crude oil price series. To determine the optimal lag order for each influencing factor series in relation to the crude oil price series, we used the Akaike information criterion (AIC) criterion (Ogasawara 2016).

3.3.2 Forecasting model

In this paper, we utilized forecasting models from previous studies with the best forecasting performance, including Long Short-Term Memory (LSTM) (Wu et al. 2021a, b), Backpropagation (BP) (Wu et al. 2021a, b), and Gated Recurrent Unit (GRU) (Jiang et al. 2022).

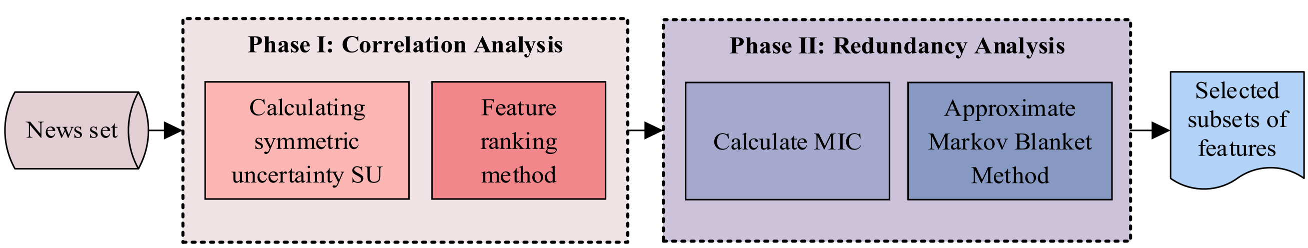

Fig. 4 Feature selection framework

1) LSTM

LSTM (Song et al. 2020) is a powerful deep-learning algorithm that includes an input layer, one or more LSTM layers, and an output layer. It is designed to learn complex, nonlinear relationships between inputs and can accommodate multiple inputs simultaneously.

2) BP

The BP (Li et al. 2012) neural network typically has a three-layer structure, including an input layer, a hidden layer, and an output layer. The model forecasting accuracy is gradually improved through the backpropagation process, whereby the connection weights and thresholds of each layer are updated and adjusted. The BP neural network is capable of changing the connection weights between artificial neurons in the forward multilayer network continuously, allowing the network to input and output the required information.

3) GRU

The GRU (Salem 2021) model preserves the influence of LSTM while simplifying its structure. Specifically, the 3-gate control mode of the forget gate, input gate, and output gate within the LSTM structure is improved into a 2-gate control mode that includes an update gate and a reset gate. This simplification of the system structure reduces the number of operations required and facilitates model training and learning.

3.3.3 Evaluation criteria

To assess the effectiveness of the forecasting models, we used three criteria: directional accuracy, mean absolute percentage error, and directional absolute error rate.

Directional accuracy (Niu et al. 2021) is a metric used to evaluate whether the predicted direction of price change by the model is consistent with the actual price change. The calculation formula for directional accuracy is as follows:

$$Direction\ accuracy = \frac{100\%}{n}\sum\nolimits_{i=1}^{n}\left(\frac{y_i - y_{i-1}}{|y_i - y_{i-1}|}\frac{\hat{y}_i - y_{i-1}}{|\hat{y}_i - y_{i-1}|}\right), \hat{y}_i \neq y_{i-1} and \hat{y}_i \neq y_i \tag{14}$$

The mean absolute percentage error (MAPE) represents the average error percentage (Hu et al. 2020) and is computed by dividing the sum of the absolute errors by the actual value. The formula for calculating MAPE is as follows:

$$MAPE = \frac{1}{n}\sum\nolimits_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y}\right| \tag{15}$$

where y_i and y_{i-1} represent the actual prices in week i and week $i-1$, respectively, and $\hat{y}_i$ represents the predicted price in week i.

In this paper, we propose a novel evaluation criterion, namely the directional absolute error rate. This criterion combines the mean absolute percentage error in the forecasting process with the accuracy of the forecasting direction to measure the effectiveness of the model's forecasting. A smaller value indicates better performance of the model.

$$R_i = \begin{cases} \frac{y_i - y_{i-1}}{|y_i - y_{i-1}|}\frac{\hat{y}_i - y_{i-1}}{|\hat{y}_i - y_{i-1}|}, \hat{y}_i \neq y_{i-1} \\ \frac{y_i - y_{i-1}}{|y_i - y_{i-1}|}1, \hat{y}_i = y_{i-1} \end{cases} \tag{16}$$

$$w_i = \begin{cases} \alpha\left|\frac{\hat{y}_i - y_i}{y_i}\right|, R_i = -1 \\ \beta\left|\frac{\hat{y}_i - y_i}{y_i}\right|, R_i = 1 \end{cases} \tag{17}$$

$$K = \frac{100\%}{n}\sum\nolimits_{i=1}^{n}(w_i) \tag{18}$$

The direction indicator R_i takes a value of 1 or −1, where 1 indicates that the predicted direction is consistent with the true direction. The error weight value w_i is also used, with a weight value of α being assigned to the incorrect forecasting direction and β to the correct direction. The K value refers to the directional absolute error rate, and a smaller K value indicates better performance of the model. In practical crude oil price forecasting, the direction of price movement is often more critical to decision-makers than the magnitude of the change. An incorrect judgment of direction, even with a small numerical error, may lead to significant losses in trading decisions, while a correct judgment of direction with a certain magnitude deviation is still valuable for capturing market trends. Therefore, it is necessary to assign a higher weight to the error when the direction is misjudged. This assignment can be adjusted according to actual circumstances.

4 Experiment study

4.1 Baselines

To comprehensively evaluate the performance of the proposed oil price fluctuation forecasting method, the ablation experiment and comparative experiment are designed in this paper.

(1) Baselines in ablation experiment

Five baselines are designed in the ablation experiment including Exp 1–5. The Exp 6 is the proposal in this paper. The detailed descriptions of the methods are described below:

Experiment 1: Forecasting using financial data and historical oil price data. This experiment employs financial data and historical oil price data to forecast oil prices.

Experiment 2: Forecasting using news sentiment feature data and historical oil price data. This experiment utilizes news sentiment feature data and historical oil price data to forecast oil prices.

Experiment 3: Forecasting using a combination of financial data, news sentiment feature data from a traditional general sentiment lexicon, and historical oil price data. This experiment combines financial data, news sentiment feature data, and historical oil price data. To compare the forecasting effects using different lexicons, this experiment uses the AFINN oil price lexicon (Nielsen 2011) to calculate sentiment values.

Experiment 4: Forecasting using a combination of financial data, news sentiment feature data from a different traditional general sentiment lexicon, and historical oil price data. This experiment combines financial data, news sentiment feature data, and historical oil price data. To compare the forecasting effects using different lexicons, this experiment uses the NRC emotion lexicon (Mohammad 2013) to calculate sentiment values.

Experiment 5: Forecasting using a combination of financial data, news sentiment feature data from a traditional oil price sentiment lexicon, and historical oil price data. This experiment combines financial data, news sentiment feature data, and historical oil price data. To compare the forecasting effects using different lexicons, this experiment uses the LM oil price lexicon to calculate sentiment values.

Experiment 6 (Our proposal): Forecasting combining financial data, news sentiment feature data from an enriched oil price forecasting sentiment lexicon, and historical oil price data. This experiment adopts the enriched oil price forecasting sentiment lexicon to calculate news sentiment values and assesses whether the lexicon constructed in this paper performs better in forecasting.

(2) Baselines in comparative experiment

This paper uses two existing state-of-the-art methods discussed in the literature review as baselines in the comparative experiment.

Cpmodel 1: Li et al., (2018) propose a crude oil price forecasting method based on online news text mining. The input data of the model is divided into two parts: news headlines and financial market data. First, the polarity score and subjectivity score are calculated for each news headline. Then, the LDA is used to classify their topics, and the topic-based features are established by average sentiment scores and a CNN classifier. Finally, a VAR model using text features and financial variables is built, and the Random Forest, Support vector regression (SVR), and Linear regression models are employed for oil price forecasting, respectively.

Cpmodel 2: Li et al., (2021) developed a crude oil forecasting approach using VMD-BiLSTM by considering investor sentiment indicators. First, a news sentiment score is established based on the cumulative effect of oil news headlines. Then, the VMD algorithm is employed to decompose the historical crude oil price series. Next, the BiLSTM, ARIMA, and SVR models are used for the oil price volatility forecasting, respectively. They also tried other two models, ARIMA and SVR.

Furthermore, to further verify the role of lexicons in sentiment computation and validate the effectiveness of the extended oil price lexicon proposed in this paper, we have added a lexicon-free sentiment computation method based on BERT as a comparative baseline.

4.2 Sentiment lexicon

In our experiment on constructing the oil price forecasting sentiment lexicon, forty-three thousand six hundred and twenty-four news articles collected from the website oilprice (https://oilprice.com) served as the corpus for lexicon development. When constructing the expanded seed lexicon LM-S within this process, we set the threshold H to 0.005 after consulting professionals. The resulting lexicon contains 896 positive words and 827 negative words, which achieves a balance between comprehensiveness and accuracy. For example, "Conflict" is included in LM-S and marked as positive because it usually causes crude oil prices to rise, while words like "Stable" that have no significant impact on oil price fluctuations are not included.

4.3 News category classification results

We labeled two thousand one hundred and twenty-five news in the website oilprice to test the performance of the news classification model. We observed class imbalance in seven categories due to the uneven number of news releases in each category. To address this issue, we employed the synthetic minority over-sampling technique (SMOTE) (Chawla et al. 2002) method to balance the imbalanced data. Table 3 presents the results of the balanced data.

Table 3 Balanced data

	Labeled data	Balanced data
OPEC + crude oil production	0:1450 1:274	0:1450 1:870
US crude oil production	0:1546 1:178	0:1546 1:927
Crude oil inventory	0:1584 1:140	0:1584 1:950
Crude oil demand	0:902 1:822	Do not need balancing
Finance	0:1137 1:587	0:1137 1: 682
Extreme weather	0:1580 1:144	0:1580 1:948
Pandemics	0:1530 1:194	0:1530 1:918
Geopolitical conflicts	0:1283 1: 441	0:1283 1:769

The news data were split into a training set and a test set using an 8:2 ratio. The classification performance of the Random Forest, LightGBM, and XGBoost models for the news categories test sets was compared using Accuracy, Precision, Recall, and F1-score, as presented in Table 4.

According to Table 4, LightGBM achieved the highest performance in the test set under all eight classifications and was therefore selected as the model for predicting classification. Among them, we have conducted a specific analysis on the phenomenon that LightGBM has a relatively low accuracy rate in the "Finance" category.

Table 5 presents the classification confusion matrix of the LightGBM model across various categories. Overall, the forecasting performance for most categories is favorable, with the majority of predicted results aligning with the actual categories. Notably, the performance of classification affects the calculation of sentiment indicators. While the F1-score for the "Finance" category is relatively low at 0.607, its accuracy rate exceeds 70%. Given that over 70% of "Finance" news is correctly classified, the core information of its sentiment features is still effectively captured.

In addition, to verify the robustness of the SMOTE method, we have added a comparative experiment of LightGBM based on ADASYN (Adaptive Synthetic Sampling) (He et al. 2008), and the relevant results are shown in Table 6.

As can be seen from the Table 6, there is no clear-cut overall superiority or inferiority between the SMOTE method and ADASYN across all categories. It should be noted that prior to conducting the experiments, we verified the

Table 4 Classification model results

	Model	Accuracy	Precision	Recall	F1-score
OPEC + crude oil production	Random Forest	0.935	0.936	0.895	0.915
	LightGBM	**0.944**	0.919	0.939	**0.929**
	XGBoost	0.940	0.914	0.934	0.924
US crude oil production	Random Forest	0.933	0.947	0.870	0.907
	LightGBM	**0.941**	0.938	0.902	**0.920**
	XGBoost	0.927	0.930	0.870	0.899
Crude oil inventory	Random Forest	0.963	0.977	0.919	0.947
	LightGBM	**0.972**	0.967	0.957	**0.962**
	XGBoost	0.939	0.933	0.898	0.915
Crude oil demand	Random Forest	0.693	0.694	0.626	0.658
	LightGBM	**0.701**	0.665	0.742	**0.701**
	XGBoost	0.672	0.662	0.626	0.644
Finance	Random Forest	0.712	0.776	0.329	0.462
	LightGBM	**0.723**	0.650	0.569	**0.607**
	XGBoost	0.695	0.635	0.445	0.524
Extreme weather	Random Forest	0.957	0.978	0.906	0.941
	LightGBM	**0.962**	0.963	0.938	**0.950**
	XGBoost	0.911	0.889	0.875	0.882
Pandemics	Random Forest	0.916	0.962	0.813	0.881
	LightGBM	**0.931**	0.932	0.882	**0.907**
	XGBoost	0.894	0.877	0.840	0.858
Geopolitical conflicts	Random Forest	0.815	0.853	0.627	0.723
	LightGBM	**0.822**	0.810	0.703	**0.753**
	XGBoost	0.800	0.788	0.658	0.717

Table 5 Confusion Matrix of LightGBM

OPEC + crude oil production		0	1
	0	268	15
	1	11	170
US crude oil production		0	1
	0	300	11
	1	18	166
Crude oil inventory		0	1
	0	315	6
	1	8	178
Crude oil demand		0	1
	0	121	61
	1	42	121
Finance		0	1
	0	185	42
	1	59	78
Extreme weather		0	1
	0	307	7
	1	12	180
Pandemics		0	1
	0	291	12
	1	22	165
Geopolitical conflicts		0	1
	0	227	26
	1	47	111

balance of data distribution across categories. The "Crude oil demand" category was found to be inherently balanced, so neither SMOTE nor ADASYN was applied to it, and thus it is not included in Table 6. Similarly, the "Finance" category was confirmed to be balanced, thus ADASYN was not used for resampling this category, so only the results of the SMOTE method for this category are shown in the Table 6. For categories that required resampling, SMOTE shows better performance in "OPEC + crude oil production" and "U.S. crude oil production" in terms of Accuracy and F1-score. For example, in "OPEC + crude oil production", SMOTE achieves an Accuracy of 0.944 and an F1-score of 0.929, both higher than ADASYN's 0.929 and 0.919 respectively. In contrast, ADASYN outperforms SMOTE in "crude oil inventory", with an Accuracy of 0.977 compared to SMOTE's 0.972 and an F1-score of 0.975 versus 0.962. This indicates that different imbalance-handling methods have their own advantages in different categories, depending on the inherent characteristics of the data.

4.4 Performance of the forecasting model

This study collected three datasets: historical crude oil price data, news text data, and financial market data. The crude oil price data consists of weekly WTI price data are collected. The news text data is a collection of news articles published on the oil price website (https://oilprice.com). The financial market data includes a total of 23 features, including U.S. Treasury yields, London gold spot prices, the Baltic dirty tanker index, active drilling, refinery starts, U.S. crude oil production, U.S. crude oil demand, U.S. crude oil inventory, WTI open interest, Brent positions, crude oil imports, Brent oil prices, the Reuters CRB index, and others, collected every week. Since all indicators are weekly frequency data, we matched and merged various financial indicators based on dates to ensure consistency in the time dimension. To verify the performance of the proposed crude oil price forecasting method, the dataset is divided into a training set and a test set for the ablation and comparative experiments: the training set for the oil price forecasting model is from November 7, 2020, to May 3, 2025, while the test set is from May 10, 2025, to July 28, 2025. A rolling window approach is adopted to estimate and test the oil price forecasting model.

Table 6 Comparison between ADASYN and SMOTE Methods (for LightGBM)

	method	Accuracy	Precision	Recall	F1-score
OPEC + crude oil production	ADASYN	0.929	0.916	0.922	0.919
	SMOTE	**0.944**	**0.919**	**0.939**	**0.929**
US crude oil production	ADASYN	0.930	0.926	**0.923**	**0.924**
	SMOTE	**0.941**	**0.938**	0.902	0.920
Crude oil inventory	ADASYN	**0.977**	**0.974**	**0.977**	**0.975**
	SMOTE	0.972	0.967	0.957	0.962
Finance	SMOTE	**0.723**	**0.650**	**0.569**	**0.607**
Extreme weather	ADASYN	0.953	0.956	**0.944**	0.949
	SMOTE	**0.962**	**0.963**	0.938	**0.950**
Pandemics	ADASYN	0.925	0.925	**0.912**	**0.918**
	SMOTE	**0.931**	**0.932**	0.882	0.907
Geopolitical conflicts	ADASYN	0.793	0.784	**0.771**	**0.776**
	SMOTE	**0.822**	**0.810**	0.703	0.753

4.4.1 Feature selection and lag order calculation

During the feature selection and dimensionality reduction process, the Symmetrical Uncertainty (SU) threshold in the correlation analysis stage was set to 0.7, retaining features that are strongly correlated with the target variable. Redundant features were eliminated by pairwise comparisons among candidate features and between each feature and the target variable. The detailed results of feature selection are shown in Table 7.

The lag order calculation was conducted on the feature-processed data, with the results shown in Table 8. The rows are the lag of weeks and the columns are the selected specific influencing factors in Table 8. The value at the intersection of rows and columns is the AIC. The row corresponding to the minimum value of AIC is the lag period of the corresponding column.

This paper utilizes three criteria to evaluate the forecasting performance of the model during the testing period, including Mean absolute percentage error (MAPE), Directional accuracy, and Directional absolute error rate. Specifically, in the experiment, after consulting experts in energy market forecasting, we determined the weights as $\alpha = 0.8$ and $\beta = 0.2$ to balance the relative importance of direction and magnitude. In addition, to measure computational complexity, two new indicators have been added: parameter count and training time. Six experiments were conducted using the LSTM, BP, and GRU methods.

The LSTM model uses the LSTM layer in Keras, python, with an input dimension of 50, a hidden layer with a dimension of 35, and a batch size of 16. The result of its hidden layer is input into a dense layer for linear transformation, and its output dimension is 35; the dropout coefficient of the above two operations is 0.3; the obtained 35-dimensional vector is then passed through a sensing layer with dimension 1 to get the oil price forecasting result of LSTM. The BP model uses sklearn's MLPRegressor with a single hidden layer, whose active function is identity, optimizer is Limited-memory Broyden-Fletcher-Goldfarb-Shannoand and penalty coefficient, alpha, is 0.001. The GRU model uses the GRU layer in Keras, python, which consists of two GRU layers and a dense layer. The output dimension of the first GRU layer is 50 with a dropout rate of 0.4; the output dimension of the second GRU layer is 30 with a dropout rate of 0.4; and the output dimension of the dense layer is 1. Both GRU model and LSTM model use Adam optimizers with a learning rate of 0.0001, and their max iteration are 200.

4.4.2 Ablation experiment results and analysis

The results of the ablation experiment are presented in Table 9.

According to Table 9, the experimental results are analyzed as follows:

(1) When comparing Experiment 1 (using financial data) and Experiment 2 (using news feature data), the performance of Experiment 2 is significantly better than that of Experiment 1 when using the GRU method. When using the LSTM and BP methods, the performance of Experiment 2 is basically better than that of Experiment 1. The direction accuracy of Experiment 2 in LSTM is lower than that of Experiment 1, while the MAPE of Experiment 2 in BP is slightly higher than that of Experiment 1. These results indicate that both financial and news feature data are effective for oil price forecasting, but their relative strengths and weaknesses vary across different models.

(2) When comparing Experiments 1 and 2 (using single-source data) to Experiments 5 and 6 (combining financial data with news sentiment features), the results of Experiments 5 and 6 were better than those of Experiments 1 and 2 using all three methods. These findings suggest that adding news sentiment features to the forecasting model is meaningful and can significantly improve forecasting performance.

(3) Comparing Experiment 5 (using the LM oil price lexicon) to Experiment 6 (using the enriched oil price forecasting sentiment lexicon), the sentiment feature indicators derived from news sentiment calculation using the lexicon proposed in this paper yielded better results on all three models. The absolute error in the direction of forecasting was greatly reduced and the directional accuracy was significantly improved using the GRU method. Meanwhile, the enriched sentiment lexicon also performed well in the other two methods. The enriched sentiment lexicon for oil price forecasting contains a larger set of oil price sentiment words and a wider variety of words, and this experimental result confirms its effectiveness in practical forecasting.

(4) Comparing Experiment 3 (using the AFINN sentiment lexicon) and 4 (using the NRC sentiment lexicon) to Experiment 6 (using the enriched oil price forecasting sentiment lexicon), the sentiment feature indicators derived from news sentiment calculation using the lexicon proposed in this paper performed the best on all three models. The enriched sentiment lexicon for oil price forecasting has more words than other common lexicons and is more relevant to the crude oil forecasting domain, and this experimental result confirms its advantage in practical forecasting.

(5) In terms of model comparison, the GRU model generally outperformed the other methods. Whether considering direction forecasting or value forecasting, GRU

Table 7 Feature selection results

ID	Specific influencing factors	Ranking by symmetrical uncertainty (SU) values	Record of Redundant Feature Removal Based on MIC Values
0	Weekly category intensity of OPEC + crude oil production	1	'Weekly category sentiment intensity of OPEC + crude oil production ', 'Weekly category sentiment intensity of US crude oil production ', 'Weekly category sentiment intensity of Crude oil demand ', 'Weekly category sentiment intensity of Finance ', 'Weekly category sentiment intensity of Crude oil inventory ', 'Weekly category sentiment intensity of Pandemics ', 'Weekly category sentiment intensity of Extreme weather '
1	Weekly category intensity of US crude oil production	1	'Weekly category intensity of Crude oil inventory ', 'Weekly category intensity of Crude oil demand ', 'Weekly category intensity of Geopolitical conflicts ', 'Weekly category intensity of Finance ', 'Weekly category intensity of Pandemics ', 'Weekly category sentiment intensity of Geopolitical conflicts ', 'Weekly category sentiment variance of US crude oil production ', 'Weekly category sentiment variance of Crude oil demand ', 'Weekly category sentiment variance of Geopolitical conflicts ', 'Weekly category sentiment variance of Finance ', 'Weekly category intensity of integration ', 'Weekly category sentiment variance of integration ', 'Weekly category sentiment variance of Pandemics ', 'Weekly category sentiment variance of Crude oil inventory ', 'Weekly category sentiment variance of Extreme weather '
2	Weekly category sentiment decay impact of OPEC + crude oil production	1	'Weekly category sentiment decay impact of US crude oil production ', 'Weekly category sentiment decay impact of Crude oil inventory ', 'Weekly category sentiment decay impact of Geopolitical conflicts ', 'Weekly category sentiment decay impact of Extreme weather ', 'Weekly category sentiment decay impact of Pandemics ', ' Crude oil inventory of OPEC + crude oil production ', 'Weekly category sentiment decay impact of integration '
3	Weekly category sentiment decay impact of Crude oil demand	1	-
4	Weekly category sentiment decay impact of Finance	1	-
5	London gold spot prices	1	'Weekly category intensity of Extreme weather ', 'the Baltic dirty tanker index'
6	WTI open interest	1	'Brent positions'
7	the Reuters CRB index	1	'U.S. crude oil inventory', 'U.S. crude oil demand', 'WTI-USG-REF', 'U.S. Treasury yields', 'refinery starts', 'U.S. crude oil production'

Table 7 (continued)

ID	Specific influencing factors	Ranking by symmetrical uncertainty (SU) values	Record of Redundant Feature Removal Based on MIC Values
8	CLc1	2	'LCOc3', 'LCOc1', 'CLc3', 'LCOc12', 'CLc12', 'CLc6', 'active drilling'
9	LCOc6	12	'crude oil imports', 'BRT-ROT-REF', 'Singapore crack spread'

Table 8 Feature lag period calculation results

AIC	Selected specific influencing factors									
ID	0	1	2	3	4	5	6	7	8	9
Lag of 0 weeks	5.926	5.8	4.01	4.042	4.048	9.865	22.663	5.466	4.183	3.8
Lag of 1 weeks	5.645	5.619	3.857	3.877	3.883	9.874	22.551	5.294	3.629	3.473
Lag of 2 weeks	5.599	5.501	3.831	3.848	3.854	9.893	22.556	5.31	3.449	3.446
Lag of 3 weeks	5.6	5.485	3.74	3.761	3.766	9.885	22.562	5.328	3.398	3.459
Lag of 4 weeks	5.61	5.501	3.705	3.729	3.73	9.921	22.586	5.35	3.401	3.449
Lag of 5 weeks	5.614	5.501	3.718	3.742	3.742	9.908	22.611	5.357	3.375	3.439
Lag of 6 weeks	5.623	5.520	3.664	3.692	3.686	9.945	22.651	5.385	3.376	3.455
Lag of 7 weeks	5.662	5.548	3.685	3.708	3.704	9.964	22.671	5.418	3.4	3.487
Lag of 8 weeks	5.632	5.511	3.707	3.727	3.726	9.975	22.675	5.407	3.404	3.491
Lag of 9 weeks	5.633	5.474	3.7	3.718	3.726	9.953	22.674	5.402	3.38	3.476
Lag period	2	9	6	6	6	0	1	1	5	5

Table 9 Comparison of experiment results

Forecasting model	Experiment	MAPE	Directional accuracy	Directional absolute error rate	Parameter count	Training time(s)
LSTM	Exp 1	0.091	0.500	0.059	18961	8.966
	Exp 2	0.080	0.375	0.056	30961	9.561
	Exp 3	0.075	0.625	0.049	30081	9.118
	Exp 4	0.077	0.375	0.058	41221	9.529
	Exp 5	0.073	0.375	0.056	32021	10.114
	Exp 6 (Ours)	0.061	0.500	0.046	32821	9.419
BP	Exp 1	0.094	0.250	0.052	3801	1.416
	Exp 2	0.095	0.500	0.047	9801	0.401
	Exp 3	0.085	0.250	0.061	12801	6.946
	Exp 4	0.085	0.125	0.067	14801	18.754
	Exp 5	0.080	0.375	0.054	10201	5.858
	Exp 6 (Ours)	0.069	0.500	0.046	10601	12.982
GRU	Exp 1	0.074	0.625	0.035	27341	3.448
	Exp 2	0.066	0.625	0.032	33821	3.465
	Exp 3	0.061	0.625	0.030	47617	5.302
	Exp 4	0.071	0.625	0.035	63771	3.321
	Exp 5	0.059	0.500	0.037	29971	3.503
	Exp 6 (Ours)	**0.043**	**0.875**	**0.02**	**30571**	**3.268**

performs better than the other two models in terms of direction accuracy, direction absolute error and MAPE value. Therefore, GRU is the best choice.

(6) In terms of computational complexity, parameter count and training time reflect the complexity from two perspectives. Comparing Experiment 1 (using financial data) with Experiment 6 (which incorporates the comprehensive oil price forecasting sentiment lexicon proposed in this study), the computational complexity of Experiment 1 is lower than that of Experiment 6 when using the LSTM and BP methods. However, when using the GRU method, there is no significant difference in computational complexity between Experiments 1 and 6. This is because Experiment 6 integrates a rich sentiment lexicon for oil price forecasting based on the financial data used in Experiment 1, which increases the number of influencing factors and thus raises the computational complexity while improving forecasting performance. Comparing Experiment 6 with Experiment 3 (using the AFINN sentiment lexicon) and Experiment 4 (using the NRC sentiment lexicon), the computational complexity of Experiment 6 shows no significant difference from that of Experiments 3 and 4 when using the LSTM and BP methods. However, when using the GRU method, Experiment 6 demonstrates lower computational complexity than Experiments 3 and 4. These results indicate that the GRU model outperforms other methods in terms of computational complexity, achieving improved forecasting performance while maintaining relatively low computational cost.

In summary, the above experimental results show that the established improved oil price forecasting sentiment lexicon and the proposed novel news sentiment indicators in this study have a significant improvement on the crude oil price fluctuation forecasting task.

4.4.3 Comparative experiment results and analysis

The comparative experimental results of two existing state-of-the-art methods and the proposal are presented in Table 10.

According to Table 10 compared to the existing state-of-the-art models, our proposed method achieves optimal results. It reduces the mean absolute percentage error (MAPE) by 24.56% and the directional absolute error rate by 39.39%, and increases the directional accuracy by 34.62% in the crude oil price fluctuation forecasting task, our proposed GRU model achieves a MAPE of 0.043, a directional accuracy of 0.875, and a directional absolute error rate of 0.020, which are all better than the metrics of non-deep learning models. Although deep learning models generally have a relatively large number of parameters and require more training time due to the complexity of their network structures and the characteristics of their training mechanisms, our proposed GRU model, despite having more parameters than BiLSTM, significantly reduces the training time while maintaining excellent forecasting performance.

The results of the comparative experiments indicate that our proposed crude oil price fluctuation forecasting method has better performance and verify its effectiveness.

We used the BERT model to perform sentiment computation on the crawled news information, and then integrated the obtained sentiment indicators with financial indicators to construct an oil price forecasting model. As a lexicon-free deep learning baseline method, the comparison results

Table 10 Comparisons with existing approaches

Models	Forecasting model	Model type	MAPE	Directional accuracy	Directional absolute error rate	Parameter count	Training time(s)
Cpmodel 1	Random Forest	Machine learning	0.057	0.625	0.037	43,813	0.113
	SVR	Machine learning	0.063	0.625	0.033	459	0.003
	Linear Regression	Traditional statistical learning	0.059	0.500	0.039	2	0.001
Cpmodel 2	ARIMA	Traditional time—series model	0.132	0.429	0.102	7	0.454
	SVR	Machine learning	0.141	0.286	0.714	12,355	0.005
	BiLSTM	Deep learning	0.407	0.650	0.350	357	219.520
Ours	**GRU**	**Deep learning**	**0.043**	**0.875**	**0.020**	**30,571**	**3.268**

Table 11 Comparison between our proposed method and BERT model forecasting

Models	MAPE	Directional accuracy	Directional absolute error rate	Parameter count	Training time(s)
BERT	0.069	0.500	0.042	64,771	3.253
Ours	**0.043**	**0.875**	**0.020**	**30,571**	**3.268**

between this model and the method based on the extended oil price lexicon proposed in this paper are shown in Table 11.

According to Table 11, compared with the baseline method based on BERT-based sentiment analysis, the proposed method based on an expanded oil price sentiment lexicon achieved better performance in the oil price forecasting task. It reduced the Mean Absolute Percentage Error (MAPE) by 26%, lowered the Directional Absolute Error Rate by 22%, and improved the Directional Accuracy by 37.5%. Moreover, while enhancing predictive performance, it also used fewer parameters and reduced computational complexity.

4.5 Explanation of forecasting results

In this section, we utilize the machine learning module coupled with the interpretable SHAP module (Xu et al. 2020) to calculate the contribution of each factor to the crude oil price. Specifically, we use the GRU model from Experiment 4, which demonstrates the best forecasting performance, to calculate the SHAP value of each feature. The SHAP values of the input features of the model are presented in Fig. 5.

At the model input level, we extended the original variables in the time dimension, introducing one lag term for each original variable (marked as *). Regarding model interpretability, we present the top 10 features that have the greatest impact on the WTI price and find that, in addition to WTI positions lag_1, five news sentiment indexes had a great impact on the current WTI price. Among these, the weekly sentiment value of US crude oil production lag_9* has the greatest impact on the current WTI price in all news sentiment index features, followed by the weekly sentiment value of OPEC + crude oil production. The results show that the news sentiment index made a significant contribution to the model output, highlighting the importance of incorporating news sentiment features in financial data forecasting, and it also explains why our model has an excellent improvement on the MAPE, the directional accuracy and the directional absolute error rate.

Based on Fig. 6, we utilize the interpretable SHAP method to calculate the contribution weights of each driving factor affecting oil price changes. The top four contributing factors are identified as the weekly sentiment value of US crude oil production (19%), the weekly category sentiment intensity of OPEC + crude oil production (16%), the weekly category sentiment intensity of finance (14%) and the LCOc6 (10%). However, we would like to highlight that the weekly category sentiment intensity of crude oil demand and the weekly sentiment value of OPEC + crude oil production also had a significant impact on oil price changes. This suggests that incorporating news sentiment indicators can provide valuable supplementary information for financial data forecasting.

5 Conclusion

This paper proposes a novel crude oil price fluctuation forecasting method that integrates news sentiment. The method can effectively enhance the performance of crude oil price fluctuation forecasting, thereby further assisting the government to formulate and adjust national economic policies and assisting business decision-makers in grasping investment options. Firstly, a rule-based approach is proposed to obtain a collection of news containing price fluctuations information. Secondly, an automatic algorithm to build an improved oil price forecasting sentiment lexicon is developed based on

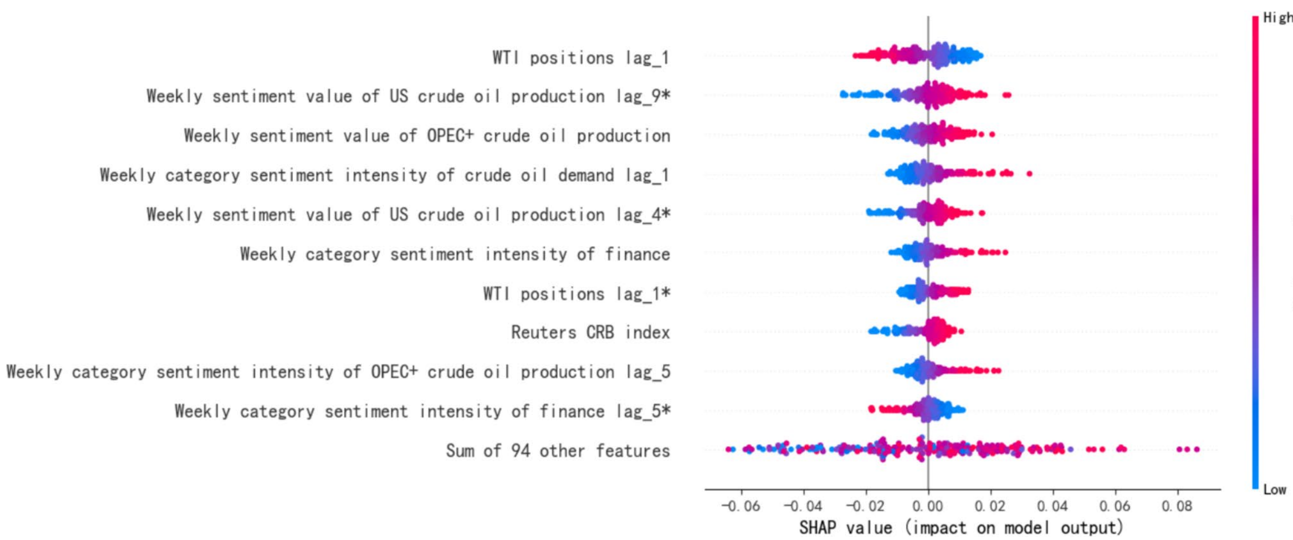

Fig. 5 Distribution of news sentiment index feature SHAP values

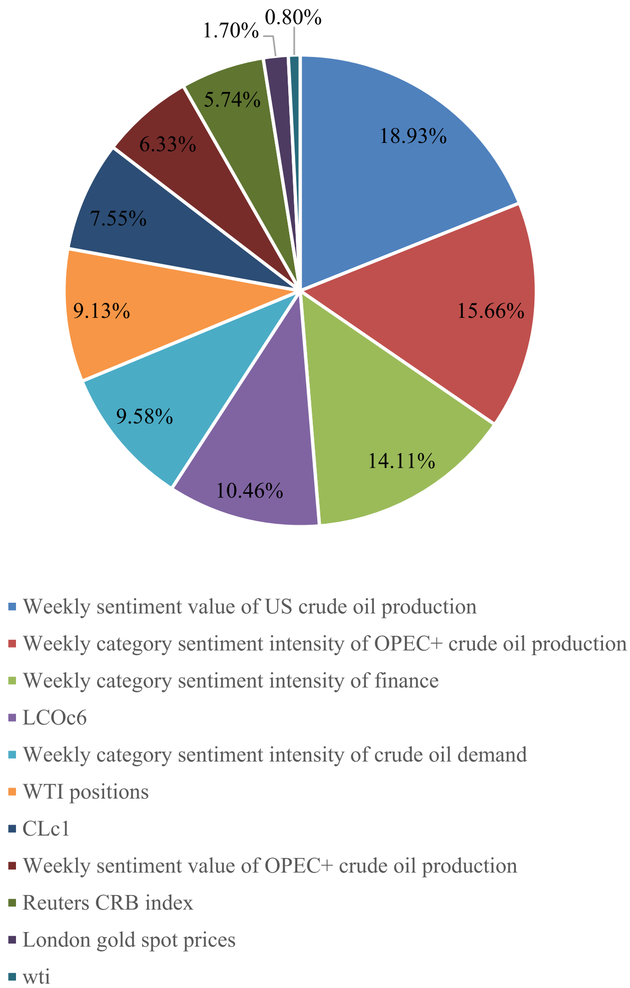

Fig. 6 Feature contribution weights

the LM financial lexicon using PMI and Word2Vec methods. Thirdly, the news is classified into eight categories and four types of sentiment indicators are designed to extract news sentiment features for forecasting. Finally, a forecasting algorithm using LSTM, BP, and GRU methods is constructed to forecast crude oil future price fluctuations, and the results are interpreted using Shapley values. Moreover, new evaluation criteria are designed to accurately measure the forecasting performance by combining the forecast values and directions. The experimental results demonstrate that our proposed model has better performance for crude oil price fluctuation forecasting and further verify its feasibility and effectiveness.

The proposed automatic algorithm for oil price forecasting sentiment lexicon can also be flexibly applied to build other domain sentiment lexicons for energy price forecasting. In future research, more online data sources can be considered, such as social media data (Dong et al. 2022b, a), to forecast crude oil price fluctuation. Meanwhile, this model will be extended to other crude oil varieties such as Brent crude oil to explore the price forecasting performance and difference rules among different crude oil categories. Additionally, regarding the resolution of data imbalance issues, we can choose a more suitable imbalance-handling method according to the specific characteristics of different categories to further improve the model's performance. Besides, future work will focus on enhancing classification accuracy, with specific efforts to reduce inter-category misclassifications as highlighted in the confusion matrix. Furthermore, future research will attempt to apply large language models to sentiment analysis in oil price forecasting, focusing on addressing issues such as unclear interpretation of sentiment scoring and inconsistent standards. Meanwhile, a comprehensive scheme suitable for the oil price domain based on large language models will also be designed to improve the performance of the oil price forecasting.

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Data availability Data will be made available on reasonable request.

Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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